

Mixed poloidal-toroidal fields in massive white dwarfs beyond barotropy

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ABSTRACT

Context. A white dwarf supported above the Chandrasekhar mass by a magnetic field needs most of that field to be toroidal, because the toroidal branch is the one that reaches the required energy. It also needs a poloidal component, or it has no exterior dipole and nothing for magnetic braking or an observer to act on.

Aims. We ask whether a self-consistent equilibrium can carry both at once, whether differential rotation then lifts it into the $1.8\text{--}2\,M_{\odot}$ range of interest for ultramassive white dwarfs, and, if not, which assumption blocks it.

Methods. We build axisymmetric non-barotropic equilibria by integrating $dP = g\,du$ along equipotentials and recovering the density from radial force balance, closing the system with a zero-temperature degenerate gas of variable composition so that the mean molecular weight per electron is diagnosed pointwise rather than imposed. Differential rotation enters through a j -constant law. We scan 908 models and accept one only if it is simultaneously in equilibrium to a virial error below 10^{-3} , below the Landau field, not shedding mass, of admissible composition on a mass-weighted pointwise test, and Ledoux-stable above the solver's own measured noise floor.

Results. The barotropic twisted torus saturates in this code at $E_{\text{tor}}/E_{\text{mag}} = 0.04$. Non-barotropy exceeds that, but the composition stratification required is Ledoux-unstable in all 126 models where it is large enough to measure. Force balance, not composition, caps the toroidal fraction near $E_{\text{tor}}/|W| \simeq 0.01$, so $B_{\varphi}/|B_p| \lesssim 0.6$ in every equilibrium, with or without rotation. Differential rotation reaches $2.85\,M_{\odot}$ in genuine equilibrium, but mass and composition trade monotonically and above $1.5\,M_{\odot}$ the solution demands $\mu_e < 2$, which no cold composition supplies. Exactly one model of 908 passes every test: $1.278\,M_{\odot}$, no toroidal field, and lighter than its own field-free background.

Conclusions. Supplying the missing pressure thermally would require $4 \times 10^9\,\text{K}$ at $1.4\,M_{\odot}$ and $2\text{--}5 \times 10^{10}\,\text{K}$ at $2\,M_{\odot}$, against a pycnonuclear carbon ignition already active at these densities. An ultramassive white dwarf supported this way does not wait: the conditions it needs are ones under which carbon has already ignited.

Key words. white dwarfs – stars: magnetic field – magnetohydrodynamics (MHD) – methods: numerical

1. Introduction

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2. What the equilibrium forces, and what barotropy adds

It is worth separating the constraints, because they are usually quoted together and only one of them is negotiable.

Consider an axisymmetric star in magnetostatic equilibrium, with no flows, and write the field through the flux function $u = \varpi A_{\varphi}$,

$$\mathbf{B} = \frac{1}{\varpi} \nabla u \times \hat{\mathbf{e}}_{\varphi} + B_{\varphi} \hat{\mathbf{e}}_{\varphi}. \quad (1)$$

Neither ∇P nor $\rho \nabla \Phi$ has an azimuthal component, so the azimuthal component of the Lorentz force must vanish by itself. That component is proportional to $(\mathbf{B}_p \cdot \nabla)(\varpi B_{\varphi})$, so

$$\varpi B_{\varphi} = \beta(u), \quad (2)$$

that is, ϖB_{φ} is constant along each poloidal field line. Outside the star the field is in vacuum and B_{φ} vanishes, so β must vanish

on every line that reaches the surface. Writing u_s for the largest value of u on the stellar surface, the toroidal field is confined to the closed-line region $u > u_s$. This is the twisted torus, and it is what our companion paper measures directly as a separatrix.

The point to make is that *none of this used barotropy*. Equation (2) and the confinement follow from axisymmetry and from the exterior being field-free. Generalising the equation of state does not release them.

What barotropy does add is a restriction on the poloidal source. If $\rho = \rho(P)$ then $\rho^{-1} \nabla P = \nabla h$ and the whole left-hand side of the Euler equation is a gradient, so $\rho^{-1} \mathbf{f}_L$ must be one too. Since the Lorentz force of Eq. (1) is everywhere parallel to ∇u , this forces $\rho^{-1} \mathbf{f}_L = M'(u) \nabla u$ for some function M , and the equilibrium collapses to the Grad–Shafranov equation (Grad & Rubin 1958; Shafranov 1966)

$$\Delta^* u = -4\pi \varpi^2 \rho M'(u) - \beta(u) \beta'(u), \quad (3)$$

with the Bernoulli integral $h + \Phi - M(u) = \text{const}$. The source of the poloidal field is pinned to the form $\rho M'(u)$: it must be proportional to the density, with a coefficient constant on flux surfaces.

That single restriction is the subject of this paper. Relaxing it is what $\rho = \rho(P, s)$ buys, because then $\rho^{-1}\nabla P$ is no longer a gradient, the baroclinic term $\nabla\rho \times \nabla P$ is available to balance what the gradient cannot, and the poloidal source becomes free (Mastrano et al. 2011; Armaza et al. 2015). It is also the route by which Cioffi & Rezzolla (2013) reach large toroidal fractions in relativistic stars, where the barotropic family had been limited to a few percent (Lander & Jones 2009; Cioffi et al. 2009; Duez & Mathis 2010).

For a white dwarf the physical case for it is straightforward: a real white dwarf is stratified, in temperature and in composition, so barotropy is the simplification and not the realism. The price is that the equilibrium no longer determines itself – a stratification has to be specified, or derived and then checked for physical plausibility.

3. The barotropic ceiling, measured

Before building a solver to beat a ceiling, the ceiling should be measured in the same code, on the same star. The value usually quoted for the barotropic twisted torus is a few percent of magnetic energy in the toroidal component; what follows is that number obtained here.

The background is a field-free configuration at $\rho_c = 10^9 \text{ g cm}^{-3}$ and $\mu_e = 2$, giving $M = 1.346 M_\odot$, $R_{\text{eq}} = 2.46 \times 10^8 \text{ cm}$ and $|W| = 2.257 \times 10^{51} \text{ erg}$. On it we solve Eq. (3) with $M'(u) = k_0$ constant, $k_0 = 10^{-13}$, and

$$\beta(u) = \beta_0 \left(\frac{u - u_s}{u_{\text{norm}}} \right)^\zeta \quad \text{for } u > u_s, \quad 0 \text{ otherwise,} \quad (4)$$

with u_{norm} fixed from the purely poloidal solution so that the functional form does not drift as u evolves. The equation is non-linear through $\beta\beta'(u)$ and is iterated to a relative change below 10^{-8} . We sweep β_0 , quoted through the field it asks for, $\beta_0 = B_{\text{ref}}\omega_{\text{ref}}$ with $\omega_{\text{ref}} = R_{\text{eq}}/2$, and repeat for $\zeta = 1, 1.1$ and 2 .

At $\beta_0 = 0$ the poloidal solution has $\max |\mathbf{B}_p| = 2.45 \times 10^{12} \text{ G}$, and its closed-line region occupies 37.7% of the stellar volume. Space is therefore not what limits the toroidal component.

The sweep is summarised in Table 1. The highest converged toroidal fraction is

$$\frac{E_{\text{tor}}}{E_{\text{mag}}} \Big|_{\text{max}} = 0.040, \quad (5)$$

at $\zeta = 1$ and $B_{\text{ref}} = 10^{12} \text{ G}$, reached after 428 iterations. Steeper β profiles do worse: $\zeta = 1.1$ converges up to 0.029 and $\zeta = 2$ only to 0.003. Above those values the iteration leaves the basin entirely rather than degrading gracefully.

Two things follow. The first is that the few-percent figure quoted from the literature is reproduced here, on a white dwarf rather than a neutron star, which is worth knowing before it is used as an argument.

The second is quantitative and much sharper. At the ceiling the toroidal field holds $E_{\text{tor}}/|W| = 1 \times 10^{-5}$. A $2 M_\odot$ configuration of the kind our companion paper certifies requires $E_{\text{tor}}/|W| = 0.203$. The barotropic twisted torus is not marginally short of what mass support needs; it is short by four orders of magnitude. Whatever supports a $2 M_\odot$ white dwarf, it is not a self-consistent barotropic mixed equilibrium.

3.1. What this measurement does not settle

Two caveats, both real.

Table 1. Converged barotropic twisted-torus configurations. Rows above the highest converged B_{ref} for each ζ are omitted because the iteration diverges rather than returning a worse equilibrium.

ζ	$B_{\text{ref}} \text{ (G)}$	$E_{\text{tor}}/E_{\text{mag}}$	$\max B_\phi \text{ (G)}$
1.0	10^{11}	0.004	9.84×10^{10}
1.0	3×10^{11}	0.029	2.97×10^{11}
1.0	10^{12}	0.040	7.23×10^{11}
1.1	3×10^{11}	0.029	3.10×10^{11}
2.0	10^{11}	0.003	1.02×10^{11}

The density is held fixed. The sweep solves Eq. (3) on the field-free background rather than iterating the stellar structure alongside it. The confinement of Sect. 2 is geometric and survives that, and at these field strengths the back-reaction on the structure is small, but the ceiling quoted here is a ceiling at fixed density and should be called that.

The failure mode is a failure of the iteration. Above the values in Table 1 an under-relaxed fixed-point scheme stops converging. That is consistent with the equilibrium ceasing to exist, and it is what the literature reports, but the two are not the same statement, and separating them needs a continuation method that follows the branch rather than a sweep that restarts at each point. Until that is done, the honest claim is that the branch is unreachable here above 4%, not that it is empty.

4. Is there room for it? The confinement and the Landau limit

Two questions decide whether the construction is worth building: whether it can give a toroidal field larger than the poloidal one, and whether it keeps the mass high. The first is about the source, which non-barotropy frees. The second is about the confinement, which it does not.

The difficulty is quantitative. The certified $2 M_\odot$ configuration carries $E_{\text{tor}}/|W| = 0.203$ with $B_\phi = K\rho\omega$, which is non-zero everywhere the density is. A mixed equilibrium must fit the same energy into the closed-line region alone, and less volume at the same energy means a larger peak field – against a configuration that already peaks at $4.28 \times 10^{13} \text{ G}$, or $0.97 B_c$ with $B_c = m_e^2 c^3 / e\hbar = 4.414 \times 10^{13} \text{ G}$. There is almost no headroom before a field-independent equation of state stops describing its own star.

We therefore ask, on the density of that configuration, what a confined $\beta(u)$ of the form (4) with $\zeta = 1$ would cost. Since E_{tor} is quadratic in β_0 , the amplitude that reproduces $E_{\text{tor}}/|W| = 0.203$ follows by scaling, and the question becomes what peak field it implies. Table 2 gives the answer over three decades of poloidal amplitude.

Three results, and the first two are favourable.

The confinement is cheaper than the space argument suggests. Reproducing $E_{\text{tor}}/|W| = 0.203$ inside 37% of the volume needs a peak of $1.58 \times 10^{13} \text{ G}$, which is $0.36 B_c$ – below the $0.97 B_c$ the space-filling $K\rho\omega$ field reaches. Confining the field does not price the mass out; it buys headroom, because the confined $\zeta = 1$ profile is flatter across its region than $K\rho\omega$ is across the star, and peak field at fixed energy is set by how peaked the profile is, not only by how much volume it has.

The ratio is a free parameter over a useful range. At the poloidal amplitude the companion paper adopts, the confined configuration has $B_t/B_p = 9.7$ in amplitude and $E_t/E_p = 884$ in energy; raising k_0 by a decade brings the amplitudes to parity, and both peaks are then still only 0.36 and 0.37 of B_c . A config-

Table 2. What a confined toroidal field costs, at fixed $E_{\text{tor}}/|W| = 0.203$, on the density of the certified $2 M_{\odot}$ configuration. Ratios B_t/B_p are of peak amplitudes, E_t/E_p of energies.

k_0	B_{pole} (G)	closed vol.	$\max B_{\varphi} $ (G)	$/B_c$	$\max \mathbf{B}_p $ (G)	$/B_c$	B_t/B_p
10^{-13}	5.9×10^{10}	37.2%	1.58×10^{13}	0.36	1.63×10^{12}	0.04	9.7
3×10^{-13}	1.8×10^{11}	37.2%	1.58×10^{13}	0.36	4.90×10^{12}	0.11	3.2
10^{-12}	5.9×10^{11}	37.2%	1.58×10^{13}	0.36	1.63×10^{13}	0.37	1.0
3×10^{-12}	1.8×10^{12}	37.2%	1.58×10^{13}	0.36	4.90×10^{13}	1.11	0.3
10^{-11}	5.9×10^{12}	37.2%	1.58×10^{13}	0.36	1.63×10^{14}	3.70	0.1

uration with comparable poloidal and toroidal fields at $2 M_{\odot}$ is not excluded by the Landau limit.

What runs out first is the poloidal side, and this is where the competition actually lives. At $k_0 = 3 \times 10^{-12}$ the poloidal peak reaches $1.11 B_c$ and the equation of state no longer describes the star it is being used on. The usable window is an exterior dipole from $\sim 6 \times 10^{10}$ to $\sim 10^{12}$ G. We had expected the competition to run through the closed-line volume instead – more escaping flux, a larger u_s , a smaller closed region – and it does not: that volume holds at 37.2% across the whole sweep, because a constant $M'(u) = k_0$ makes the Grad–Shafranov source linear in k_0 , so u and u_s scale together and the geometry is invariant.

4.1. What this does not show

It is a scoping calculation, on a fixed density, and it establishes only that the energy budget and the field strengths are mutually compatible. It does not exhibit an equilibrium. The argument that the mass survives rests on the virial relation containing E_{mag} as a whole rather than its distribution, so that a redistribution of the same magnetic energy leaves the global balance unchanged – which is sound to first order and is exactly what the next section has to verify rather than assume.

5. The non-barotropic construction

5.1. Equilibrium without a barotrope

Dropping $\rho = \rho(P)$ removes the requirement that $\rho^{-1} \mathbf{f}_L$ be a gradient, but not the requirement that the Lorentz force be parallel to ∇u , which follows from Eq. (1) alone. We may therefore write

$$\mathbf{f}_L = g \nabla u, \quad g = -\frac{\Delta^* u + \beta \beta'}{4\pi \varpi^2}, \quad (6)$$

where g is now a function of position and not of u alone. The Euler equation becomes

$$\nabla P + \rho \nabla \Phi = g \nabla u. \quad (7)$$

Taking the component along a surface of constant Φ gives

$$dP = g du \quad \text{along equipotentials}, \quad (8)$$

which determines P everywhere once it is fixed on one curve crossing every equipotential; we integrate outward along equipotentials from the polar axis, where we impose $P(\text{axis}) = \alpha P_{\text{ff}}(\text{axis})$ with P_{ff} the pressure of the field-free star at the same central density. The dimensionless anchor α is a free parameter of the construction and, as Sect. 8 shows, it is the knob that controls the composition the solution demands. The radial component of Eq. (7) then returns the density,

$$\rho = \frac{g \partial_r u - \partial_r P}{\partial_r \Phi}, \quad (9)$$

and Φ is recovered from ρ by solving Poisson's equation. The three steps are iterated to convergence.

The logic is worth stating plainly, because it is what the rest of the paper tests. The poloidal source is now free: we may choose $\Delta^* u$ as we like, and the pressure adjusts. Nothing is determined by an equation of state, because no equation of state has been imposed. What the solution owes us instead is a demonstration that the (P, ρ) pair it produces corresponds to matter that could exist.

5.2. The composition closure, and what it can and cannot absorb

We close the system with a zero-temperature degenerate electron gas of variable composition. Then $P = Af(x)$ with A independent of the mean molecular weight per electron, while $\rho = B(\mu_e) x^3$ with $B \propto \mu_e$. The relativity parameter x is therefore fixed by P alone, and

$$\mu_e \propto \frac{\rho}{x^3(P)}, \quad (10)$$

so that μ_e is not an input but a diagnostic: it is read off pointwise as the mismatch between the density and the pressure the solution carries. Less pressure supporting a given density means a larger μ_e , and more pressure means a smaller one.

This is the precise sense in which non-barotropy is composition stratification here. Fixing μ_e at a single value everywhere would restore $P = P(\rho)$ and return us to Sect. 3. The freedom that lets the construction exceed the barotropic ceiling is exactly the freedom in μ_e , and the question the construction must answer is whether the μ_e field it asks for is one a white dwarf can supply.

The two directions are not symmetric, and this asymmetry turns out to decide the result.

- $\mu_e > 2$ is ordinary. Neutron-rich species supply it: ^{56}Fe gives 2.154, ^{22}Ne gives 2.2. A cold interior can be built with them.
- $\mu_e < 2$ has no cold realisation at all. Every α -chain nucleus — ^4He , ^{12}C , ^{16}O , ^{20}Ne , ^{24}Mg — has $A/Z = 2$ exactly, and going below requires hydrogen, which is not available in the interior of an ultramassive white dwarf.

A solution demanding $\mu_e < 2$ is therefore not asking for a composition. By Eq. (10) it is asking for more pressure than cold degenerate matter provides at that density, and the only physical source of that excess is finite temperature. We return to this in Sect. 9, where it becomes the decisive constraint.

We accept $\mu_e \in [2.0, 2.15]$ as the strict window and report $[2.0, 2.2]$ separately, since ^{22}Ne is a defensible contaminant of a C/O interior.

5.3. The toroidal component

Equation (2) and the confinement to $u > u_s$ survive the loss of barotropy, as Sect. 2 established, so we keep the form of Eq. (4)

and normalise β_0 to a target $E_{\text{tor}}/|W|$. Two details of the implementation are not cosmetic, and we record them because each one produced a converged but spurious answer before it was fixed.

First, u_s must be recomputed against the current star at every outer iteration, not held at the field-free background. Freezing it left the toroidal field confined to the region belonging to a $1.74 M_\odot$ configuration while the iteration converged on a $1.21 M_\odot$ one, so part of the field lay outside the matter with nothing to confine it. The resulting virial error was 0.17 and was stable from 32 to 256 outer iterations: converged, and converged on something that was not an equilibrium.

Second, $\zeta = 2$ rather than $\zeta = 1$. At $\zeta = 1$ the derivative β' jumps across the separatrix, which is a current sheet. Localising the residual of Eq. (7) showed it piled up in the equatorial plane at $0.4\text{--}0.6 R_{\text{eq}}$ — on the separatrix — and carried essentially the whole virial error; raising ζ to 2, so that β and β' both vanish there, cut the error at $E_{\text{tor}}/|W| = 0.10$ from 0.029 to 0.010. This is the same choice made by Cioffi & Rezzolla (2013) and Lander & Jones (2009), and for the same reason.

5.4. Differential rotation

We add rotation on the j -constant law,

$$\Omega(\varpi) = \Omega_c \frac{A^2}{A^2 + \varpi^2}, \quad \Psi(\varpi) = -\frac{\Omega_c^2 A^2}{2} \frac{\varpi^2}{A^2 + \varpi^2}, \quad (11)$$

with A quoted as a fraction of the background equatorial radius; $A \rightarrow \infty$ is uniform rotation. Because Ω depends on ϖ alone the centrifugal force is conservative, and the whole construction goes through with Φ replaced by $\Phi_{\text{eff}} = \Phi + \Psi$ in Eqs. (8) and (9). Gravity itself is still solved from ρ alone. We normalise Ω_c to a target $T/|W|$, renormalising against the current $|W|$ at each iteration so that the quoted ratio is the one attained at the solution and not at the starting guess.

6. Numerical method, and what the acceptance tests cost

The meridional grid is 129×129 in (r, θ) with Poisson and Grad–Shafranov solved to $l_{\text{max}} = 16$, and 32 outer iterations; doubling either the grid or the iteration count moves the reported virial errors by less than one part in 10^2 of their own value.

A configuration is accepted only if it passes, simultaneously:

Equilibrium. The virial error $\text{VE} = |2T + W + 3 \int P dV + E_{\text{mag}}|/|W| < 10^{-3}$. This is not a formality. We verified each term separately against its own identity — $\int \mathbf{x} \cdot \rho \nabla \Phi dV = -W$ to 1 part in 10^4 , $\int \mathbf{x} \cdot \nabla P dV = -3 \int P dV$ to 7 parts in 10^4 , and $\int \mathbf{x} \cdot \mathbf{f}_L dV = E_{\text{mag}}$ to 1.3 parts in 10^3 — and confirmed that VE equals the integrated radial force imbalance. A configuration with $\text{VE} = 0.26$ has, in the worst 5% of its volume, a pointwise force residual of 0.35 of the local gravity, against 1.7×10^{-4} for an accepted one.

Composition. Mass-weighted percentiles $\mu_e^{(1)}$ and $\mu_e^{(99)}$ inside the window, and at most 1% of the mass outside it. The gate must be pointwise. A shell-averaged version reported $\mu_e \in [2.048, 2.630]$ for a configuration whose pointwise maximum was 65.2, and reported $[2.002, 2.122]$ — inside the strict window — for one whose pointwise maximum was 2.627. Mass weighting is what handles the stellar surface, where the inversion of Eq. (10) is ill-conditioned as $P \rightarrow 0$:

that shell holds almost no mass and cannot move a mass-weighted percentile, so no *ad hoc* radial cut is needed.

Stratification. The composition profile must be measurable above the solver's own noise, and Ledoux-stable. We calibrate the noise by running the construction at $k_0 = 0$, where the answer must be homogeneous; the residual spread is 0.0203, and we require a profile at least $3\times$ that before computing a Ledoux criterion at all. Profiles below that threshold are not declared stable, they are declared unmeasured.

Landau. $\max |\mathbf{B}| < B_c = 4.414 \times 10^{13} \text{ G}$, applied to the total field.

Core support. The mass-weighted radial Lorentz force inside $0.2 R$, as a fraction of local gravity, must be positive.

Rotation. No mass shedding, and the attained $T/|W|$ within 5% of its target. The second test matters: the iteration runs away at large $T/|W|$ with concentrated shear, and without a guard it returned $M = 2 \times 10^{13} M_\odot$ with $\text{VE} = 1705$ as though it were a measurement. Of the 780 rotating models attempted, 334 were rejected this way.

7. Results without rotation

We scanned 128 non-rotating models over $\rho_c \in \{10^9, 3 \times 10^9\} \text{ g cm}^{-3}$, $k_0 \in [1.0, 2.5] \times 10^{-12}$, $\alpha \in [0.70, 1.00]$ and $E_{\text{tor}}/|W| \in \{0, 0.005, 0.01, 0.02\}$.

Not one is admissible. The individual pass rates are 104/128 on equilibrium, 48/128 on the Landau limit, 128/128 on core support, and 2/128 on composition; the intersection is empty.

Three results are worth separating out.

The toroidal field does not add mass here. Along any family at fixed (k_0, α, ρ_c) the mass falls monotonically as the toroidal component is turned up: $1.407, 1.386, 1.366$ and $1.279 M_\odot$ at $E_{\text{tor}}/|W| = 0, 0.005, 0.01$ and 0.02 . What rises with it is μ_e , and the rise is localised: the maximum of μ_e migrates from $0.98 R$ — a surface artefact of the ill-conditioned inversion — to $0.49 R$ in the equatorial plane, which is where $|B_\phi|$ peaks, at $0.48 R$. The toroidal field takes over part of the support locally, and Eq. (10) records that as a local demand for heavier composition.

Equilibrium itself limits the toroidal fraction. The virial error crosses 10^{-3} between $E_{\text{tor}}/|W| = 0.01$ and 0.02 . In consequence $B_\phi/|B_p|$ never exceeds about 0.6 in any model that is in equilibrium. The toroidal-dominated interior is not reached.

The stratification is Ledoux-unstable. Of the 126 models whose composition profile rises above the noise floor, *every one* is Ledoux-unstable, 43 of them maximally so. The two models that pass the Ledoux test do so only because their stratification falls below the threshold and is therefore unmeasured, and both of those fail composition. The two models with admissible composition are both purely poloidal, at 1.262 and $1.336 M_\odot$, and both are maximally Ledoux-unstable.

This last point is the mechanism, and it is not a numerical accident. The extra support that non-barotropy buys is bought by a composition gradient, and the gradient it needs has the heavy material on the outside.

8. Results with differential rotation

We then scanned 780 models adding $T/|W| \in \{0, 0.02, 0.04, 0.06, 0.08\}$ and $A/R_{\text{eq}} \in \{0.5, 1.0, 2.0\}$, and

Table 3. Mass against the composition it demands, over the 213 rotating models in equilibrium, below B_c , on their rotation target and not shedding. $\mu_e^{(1)}$ is the mass-weighted first percentile; the last column is the pressure excess over cold $\mu_e = 2$ matter that the median value implies.

$M (M_\odot)$	n	median $\mu_e^{(1)}$	implied $\Delta P/P$
< 1.3	40	2.045	—
1.3–1.5	48	1.912	6%
1.5–1.8	61	1.780	16%
1.8–2.2	38	1.549	42%
2.2–3.0	26	1.363	62%

extending the pressure anchor down to $\alpha = 0.40$; 334 ran away and were rejected, leaving 446.

Mass stops being the difficulty. Rotation delivers it. The heaviest model in genuine equilibrium reaches $2.851 M_\odot$ at $T/|W| = 0.080$, with $VE = 2.4 \times 10^{-4}$, $B < 0.56 B_c$, no mass shedding, and its core held. This is a different situation from the barotropic toroidal branch, which reached $2 M_\odot$ and then collapsed dynamically within a few dynamical times.

Mass and composition trade against each other, monotonically. Table 3 gives the median $\mu_e^{(1)}$ against mass over the models that are in equilibrium, below B_c , on their rotation target and not shedding. There is no elbow and no island: every increment of mass is paid for in μ_e , and the payment runs the wrong way — towards $\mu_e < 2$, the direction Sect. 5.2 showed has no cold realisation.

Rotation does not lift the toroidal ceiling. At $T/|W| = 0.04$ the virial error runs 1.2×10^{-4} , 3.2×10^{-3} , 1.7×10^{-2} and 0.26 at $E_{\text{tor}}/|W| = 0.005$, 0.02, 0.05 and 0.10. The gate is crossed in the same place as without rotation; the ceiling moves from about 0.01 to about 0.015, which is not a qualitative change. The $2.10 M_\odot$ model that appears at $E_{\text{tor}}/|W| = 0.10$ has $VE = 0.26$ and is not an equilibrium.

One model is admissible, and it is unremarkable. Nothing passes every test with the strict window. Widening it to $\mu_e \leq 2.2$, which ^{22}Ne justifies, admits exactly one model out of 780:

$$M = 1.278 M_\odot, \rho_c = 3 \times 10^9 \text{ g cm}^{-3}, \alpha = 0.55, T/|W| = 0.020, A = R_{\text{eq}}, B_\varphi/|B_p| = 0, \mu_e \in [2.020, 2.196] \text{ with } 0.5\% \text{ of the mass outside, Ledoux } 0.259, E_{\text{mag}}/|W| = 0.011, B = 0.81 B_c, VE = 5.3 \times 10^{-4}.$$

It is the first fully certified non-barotropic configuration we have obtained, and it carries no toroidal field, sits below the Chandrasekhar mass, and — the point that should not be buried — is *lighter* than its own field-free background, 1.278 against $1.387 M_\odot$. The construction that was meant to add support subtracted 8% of the mass. Since $\alpha < 1$ lowers the central pressure, this is self-consistent; but it means that not one admissible model in this paper is held up by its field.

We note that the 1% mass-fraction test was evaluated against the strict window even when the percentile window was widened, so it is stricter than the widened window and the count of one is a lower bound.

Table 4. Temperature that would have to supply the missing pressure, at $\rho_c = 3 \times 10^9 \text{ g cm}^{-3}$ where $P_{\text{deg}} = 2.12 \times 10^{27} \text{ erg cm}^{-3}$.

μ_e demanded	$\Delta P/P$	T (K)
1.945	3.8%	3.9×10^9
1.90	7.1%	7.2×10^9
1.70	24%	2.5×10^{10}
1.50	47%	4.8×10^{10}

9. What the composition demand means physically

Every high-mass model asks for μ_e well below 2, and Sect. 5.2 showed that this is a request for pressure, not for composition. In the ultrarelativistic limit $P \propto \mu_e^{-4/3}$, so the excess required is $(2/\mu_e)^{4/3} - 1$. Supplying it as ideal ion thermal pressure of ^{12}C , $P_{\text{ion}} = \rho kT/(Am_u)$, fixes the temperature in Table 4.

At $\rho_c = 3 \times 10^9 \text{ g cm}^{-3}$ carbon is already in the pycnonuclear regime, where C+C proceeds without thermal assistance at all. Requiring in addition a temperature of several times 10^9 K puts the configuration an order of magnitude above any carbon ignition threshold, and the $1.8\text{--}2.2 M_\odot$ band of Table 3 requires $2\text{--}5 \times 10^{10} \text{ K}$, which is photodisintegration and NSE territory rather than white dwarf territory.

The conclusion is not that the thermal route is unexplored but that it is self-defeating: a configuration in this mass range, supported this way, would have ignited before it could be assembled. We emphasise that this is an order-of-magnitude argument with an ideal ion gas and not a solved finite-temperature equation of state; radiation pressure is subdominant at these densities. The margin, however, is a factor of 5 to 100 above ignition, so the conclusion does not rest on the approximation.

It is worth adding that every realistic refinement of the equation of state moves in the same, unhelpful direction. Coulomb and lattice corrections lower the pressure by one to two percent here; general relativity lowers the maximum mass; Landau quantisation softens the equation of state. None of them supplies the 24% that $1.8 M_\odot$ needs. The Landau limit, incidentally, is not what closes the route: the certified model sits at $0.81 B_c$, comfortably below.

10. Conclusions

We set out to determine whether abandoning barotropy allows a white dwarf to carry a strong mixed poloidal–toroidal field, and, with differential rotation added, to reach the $1.8\text{--}2 M_\odot$ range of interest for ultramassive white dwarfs. It does not, and the obstruction is specific enough to be useful.

1. Measured in this code, the barotropic twisted torus saturates at $E_{\text{tor}}/E_{\text{mag}} = 0.04$ and $E_{\text{pol}}/|W| = 0.018$ (Sect. 3).
2. The non-barotropic construction can exceed that, but the composition stratification it requires to do so is Ledoux-unstable in every model where the stratification is large enough to be measured at all — 126 of 126 (Sect. 7).
3. Equilibrium, not composition, limits the toroidal fraction: the virial error crosses 10^{-3} near $E_{\text{tor}}/|W| \approx 0.01$, and $B_\varphi/|B_p| \lesssim 0.6$ in every model that is in equilibrium, with or without rotation. A toroidal-dominated interior was not obtained.
4. With differential rotation, mass ceases to be the difficulty — $2.85 M_\odot$ in genuine equilibrium — but mass and composition trade monotonically, and above $1.5 M_\odot$ the required μ_e falls below 2, which no cold composition supplies (Table 3).

5. Supplying that deficit thermally requires 4×10^9 K at $1.4 M_{\odot}$ and $2\text{--}5 \times 10^{10}$ K at $2 M_{\odot}$, against a pycnonuclear carbon ignition that is already active at these densities (Sect. 9).
6. Of 908 models across both scans, exactly one satisfies every test, at $1.278 M_{\odot}$, with no toroidal field, and lighter than its own field-free background.

The astrophysical reading is that an ultramassive white dwarf held up in this way does not wait. The support it would need corresponds to conditions under which carbon has already ignited, which bears directly on the question of whether such objects end as thermonuclear supernovae or collapse, and answers it from the side of the equilibrium structure rather than from the evolution.

What would change the answer is a genuinely self-consistent mixed Grad–Shafranov solve, in which u responds to β instead of being taken from a purely poloidal solution. We expect that to raise the toroidal ceiling of point 3, since that ceiling is a failure of force balance rather than of composition. We do not expect it to help with points 4 and 5, and it will probably make them worse, since the composition demand grows with field strength.

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